ENERGY SYSTEMS, CHARGING METHODS, AND POWER ARCHITECTURE EVOLUTION

General Power Architecture Concept

The longitudinal penile physiology monitoring device may incorporate a variety of power architectures configured to support extended wearable operation while maintaining minimal device profile and user comfort. Power systems may include rechargeable energy storage, replaceable power sources, hybrid energy configurations, or distributed energy systems depending on device embodiment.

Energy management may prioritize long-duration physiologic monitoring through adaptive sampling control, duty cycling of sensing components, and dynamic allocation of computational resources based on operational mode.

Rechargeable and Replaceable Energy Storage

Energy storage elements may include lithium-based rechargeable batteries, solid-state batteries, thin-film batteries, flexible battery structures, supercapacitors, hybrid capacitor-battery systems, or other electrical energy storage technologies suitable for wearable applications. In certain embodiments, replaceable batteries may be used to simplify maintenance or extend operational lifetime.

Charging and Energy Transfer Methods

Charging methods may include wired electrical charging, magnetic connector systems, inductive wireless charging, resonant wireless power transfer, conductive docking interfaces, or other energy transfer mechanisms. Charging systems may support sealed waterproof designs or contactless recharging to improve durability and hygiene.

Energy transfer may occur while the device is removed from the body or, in certain embodiments, during intermittent wear depending on safety and operational configuration.

Energy Harvesting Embodiments

Future embodiments may incorporate energy harvesting technologies capable of supplementing stored energy. Harvesting mechanisms may include motion-based generators, thermoelectric harvesting from body heat, piezoelectric energy conversion, ambient radio-frequency energy capture, photovoltaic elements, or biomechanical energy recovery systems.

Energy harvesting may operate independently or in combination with rechargeable power systems to extend operational lifetime between charging cycles.

Adaptive Power Management

Power management circuitry may dynamically adjust voltage regulation, processor performance, sensor activation schedules, wireless communication intervals, and data processing intensity based on detected physiologic activity or user-selected operating modes. Predictive power budgeting algorithms may optimize energy usage across extended monitoring sessions.

The power architecture embodiments described herein are illustrative and non-limiting and encompass present and future energy systems capable of supporting wearable longitudinal physiologic monitoring.